

Quantiles and Ironing

Recap

Myerson's theory for independent single-parameter values and normalized DSIC payments says: expected revenue is equal to expected virtual welfare $\mathbb{E}_{\vec{v} \sim \vec{F}}[\sum_i \varphi_i(v_i)x_i(\vec{v})]$. Then the allocation rule that maximizes this is, for each $\vec{v}$, to allocate to the bidder with the highest non-negative virtual value, using a fixed priority order to break ties. This allocation will be truthful if it is monotone, which is guaranteed when $\varphi_i(v_i)$ is monotone non-decreasing in v_i , which occurs exactly when the distribution F_i is regular. If this is the case, we can apply the payment identity to the allocation rule and this yields a DSIC mechanism.

Quantile Space

Assume F is absolutely continuous on $[0, v_{\max}]$, with positive density on the interior and finite $v_{\max}$. In *value space*:

- an agent has value v .
- the fraction of the distribution with value above v is $1 - F(v)$.
- the revenue from posting a “take-it-or-leave-it” price of v is $v[1 - F(v)]$.

In *quantile space*: $q = 1 - F(v)$.

- an agent has value v .
- the fraction of the distribution with value above v is $q(v) = 1 - F(v)$.
- the revenue from posting a “take-it-or-leave-it” price of $v(q) = F^{-1}(1 - q)$ is $v(q) \cdot q$.

Example: Consider a distribution that is $U[\$0, \$10]$. Then the quantile 0.1 corresponds to \$9, where 10% of the population might have a higher value. We let $v(q)$ denote the corresponding value, so $v(0.1)$ is \$9.

Definition 1. The *quantile* of a single-dimensional agent with value $v \sim F$ is the measure with respect to F of stronger values, i.e., $q = 1 - F(v)$; the inverse demand curve maps an agent's quantile to their value, i.e., $v(q) = F^{-1}(1 - q)$.

Quantile Distribution: Quantiles are particularly useful because we can draw an agent from any distribution by drawing a quantile $q \sim U[0, 1]$. That is, for any $\hat{q}$ and any such distribution F , $\Pr_F[q \leq \hat{q}] = \hat{q}$. In English: the probability that an agent has a value in the top 0.3 of the distribution is 0.3.

Note: For everything we do here, we *could* stay in value space, (and sometimes we'll compare), but we'd have to normalize by the distribution using $f(v)$, which makes everything a bit messier and a bit trickier.

Example: For the example of a uniform agent where $F(z) = z$, the inverse demand curve is $v(q) = 1 - q$.

For an allocation rule $x(\cdot)$ in value space, we define an allocation rule in *quantile space* $y(\cdot)$:

$$y(q) = x(v(q)).$$

Holding any other bids fixed, as $x(\cdot)$ is monotone weakly increasing, then $y(\cdot)$ is monotone *weakly decreasing*.

Definition 2. The *revenue curve* of a single-dimensional agent in value space is specified by $R(v) = v \cdot [1 - F(v)]$.

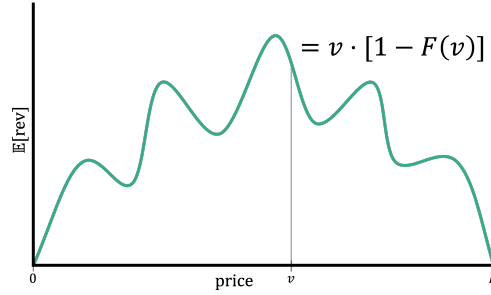


Figure 1: A revenue curve in *value space*.

Note: This is *only* the revenue that can be achieved by posting a single take-it-or-leave-it price. This does not capture the expected revenue of any given mechanism.

Definition 3. The *revenue curve* of a single-dimensional linear agent specified by inverse demand curve $v(\cdot)$ is $P(q) = q \cdot v(q)$ for any $q \in [0, 1]$.

Assuming the lower-end of the support of F is 0 and the upper end is some finite $v_{\max}$, then $P(0) = 0$ and $P(1) = 0$.

Claim 1. Any weakly decreasing allocation rule $y : [0, 1] \rightarrow [0, 1]$ can be expressed as a distribution of posted prices, allowing price zero and a never-serve option.

Proof. Given the allocation rule $y(\cdot)$, take its right-continuous version and consider the distribution $G^y(z) := 1 - y(z)$ for $0 \leq z < 1$, with $G^y(1) = 1$. Mass at quantile 0 represents never serving, and mass at quantile 1 represents price zero; acceptance at equality handles the endpoint types and jumps. We show that the mechanism that randomly draws a quantile $\hat{q} \sim G^y$ from the distribution G^y and posts the price $v(\hat{q})$ is equivalent.

For a random price $v(\hat{q})$ and fixed interior quantile q , then

$$\Pr_{\hat{q} \sim G^y}[v(\hat{q}) < v(q)] = \Pr_{\hat{q} \sim G^y}[\hat{q} > q] = 1 - G^y(q) = y(q). \quad \square$$

Claim 2. Any DSIC allocation rule $x(\cdot)$ can be expressed as a distribution of posted prices, allowing price zero and a never-serve option.

See Figure 2 for an example. When x is continuously differentiable, the interior PDF of the distribution of randomized prices is $x'(v)$ for a price of v to achieve allocation rule x . Jumps in x correspond to point masses on prices.

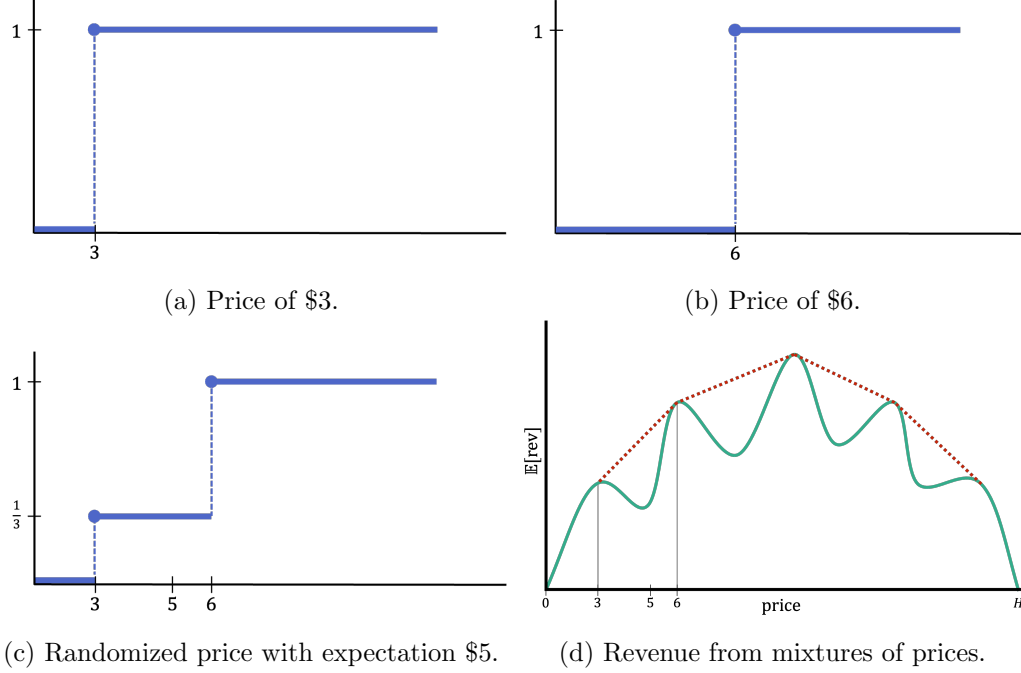


Figure 2: (a) An allocation rule for a take-it-or-leave-it price of \$3. (b) An allocation rule for a take-it-or-leave-it price of \$6. (c) An allocation that can be written $x(v) = 0$ for $v < 3$, $x(v) = \frac{1}{3}$ for $v \in [3, 6)$, and $x(v) = 1$ for $v \geq 6$. Alternatively, a randomized take-it-or-leave-it price that is \$3 with probability $\frac{1}{3}$ and \$6 with probability $\frac{2}{3}$, that is, $\$5 = \frac{1}{3} \cdot 3 + \frac{2}{3} \cdot 6$ in expectation. (d) The revenue curve in value space, including revenues from mixtures of prices indexed by their mean price. A mixture need not have the same revenue as posting its mean as a single price.

Back to Quantile Space and Ironing

Claim 3. A distribution F is regular if and only if its corresponding revenue curve in quantile space is concave.

Observe that $P'(q) = \varphi(v(q))$:

$$P'(q) = \frac{d}{dq} (q \cdot v(q)) = v(q) + qv'(q) = v - \frac{1 - F(v)}{f(v)} = \varphi(v(q)).$$

Writing $\varphi(q)$ for $\varphi(v(q))$ in quantile space, $\Phi(q) = \int_0^q \varphi(\hat{q}) d\hat{q} = P(q)$.

To summarize: a distribution F is regular if and only if:

- its corresponding revenue curve *in quantile space* is concave.
- $\varphi(v)$ is weakly increasing in value, or equivalently $\varphi(q)$ is weakly decreasing in quantile.

Definition 4. The *ironing procedure* for (non-monotone) virtual value function φ (in quantile space) is:

- Define the cumulative virtual value function as $\Phi(\hat{q}) = \int_0^{\hat{q}} \varphi(q) dq$.
- Define ironed cumulative virtual value function as $\bar{\Phi}(\cdot)$ as the least concave function lying weakly above $\Phi(\cdot)$ (its upper concave hull).
- Define the ironed virtual value function as $\bar{\varphi}(q) = \frac{d}{dq} \bar{\Phi}(q) = \bar{\Phi}'(q)$.

Summary: Write $\bar{P} = \bar{\Phi}$. Take the concave hull of the revenue curve in quantile space. Its derivative forms the ironed virtual values. (The derivatives of the original curve are the original virtual values.)

For the bimodal example from L4,

$$v \sim \begin{cases} U[0, 3] & \text{w.p. } \frac{3}{4}, \\ U(3, 8] & \text{w.p. } \frac{1}{4}, \end{cases} \quad f(v) = \begin{cases} \frac{1}{4} & v \in [0, 3], \\ \frac{1}{20} & v \in (3, 8], \end{cases} \quad \varphi(v) = \begin{cases} 2v - 4 & v \in [0, 3], \\ 2v - 8 & v \in (3, 8]. \end{cases}$$

Thus $\varphi(3.5) = -1 < \varphi(2.5) = 1$: maximizing virtual welfare can stop serving a bidder as their value increases.

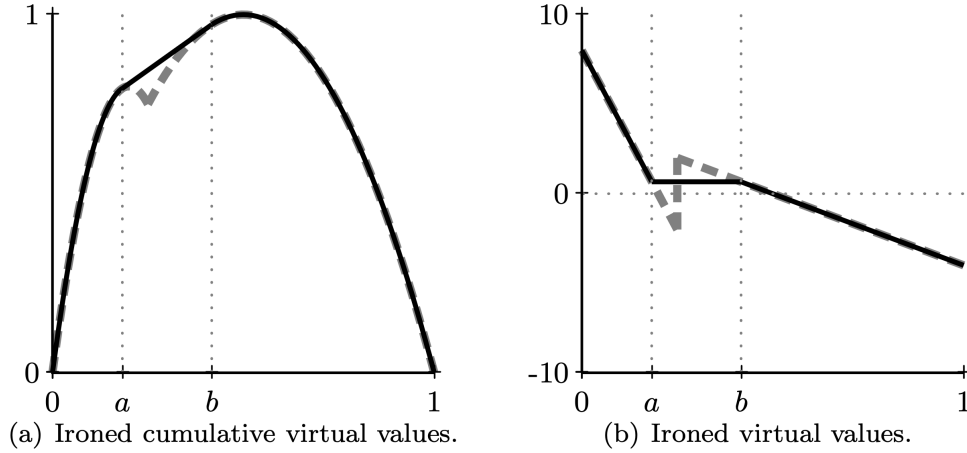


Figure 3: The bimodal agent's (ironed) revenue curve and virtual values in quantile space.

Theorem 1. For any monotone allocation rule $y(\cdot)$ and the virtual value function $\varphi(\cdot)$ of F , the expected virtual surplus of an agent is upper-bounded by their expected ironed virtual surplus, i.e.,

$$\mathbb{E}[\varphi(q)y(q)] \leq \mathbb{E}[\bar{\varphi}(q)y(q)].$$

Furthermore, this inequality holds with equality if the allocation rule y is constant on each open interval where $\bar{\Phi}(q) > \Phi(q)$.

Proof. First suppose y is continuously differentiable. Recall integration by parts:

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx.$$

By integration by parts for the virtual value function $\varphi(\cdot)$ of F and monotone allocation rule $y(\cdot)$, with expectations over $q \sim U[0, 1]$,

$$\mathbb{E}[\varphi(q)y(q)] = \mathbb{E}[-y'(q)\Phi(q)].$$

Step by step, that is,

$$\begin{aligned} \mathbb{E}[\varphi(q)y(q)] &= \int_0^1 \varphi(q)y(q) dq & q \sim U[0, 1] \\ &= \Phi(1)y(1) - \Phi(0)y(0) - \int_0^1 y'(q)\Phi(q) dq \\ &= 0 + \mathbb{E}[-y'(q)\Phi(q)]. \end{aligned}$$

because $\Phi(1) = 1 \cdot v(1) = 0$ as $v(1) = 0$, and $\Phi(0) = 0 \cdot v(0) = 0$. Notice that the weakly decreasing monotonicity of the allocation rule $y(\cdot)$ implies the non-negativity of $-y'(q)$. With the left-hand side of equation as the expected virtual surplus, it is clear that a higher cumulative virtual value implies no lower expected virtual surplus. By definition of $\bar{\Phi}(\cdot)$ as the concave hull of $\Phi(\cdot)$, $\Phi(q) \leq \bar{\Phi}(q)$ and, therefore, for any monotone allocation rule, in expectation, the ironed virtual surplus is at least the virtual surplus, i.e., $\mathbb{E}[-y'(q)\Phi(q)] \leq \mathbb{E}[-y'(q)\bar{\Phi}(q)]$.

To see the equality under the assumption that $y'(q) = 0$ for all q where $\bar{\Phi}(q) > \Phi(q)$, rewrite the difference between the ironed virtual surplus and the virtual surplus as,

$$\mathbb{E}[\bar{\varphi}(q)y(q)] - \mathbb{E}[\varphi(q)y(q)] = \mathbb{E}[-y'(q)(\bar{\Phi}(q) - \Phi(q))].$$

The assumption on y' implies the term inside the expectation on the right-hand side is zero $\forall q$. For a general monotone allocation, replace $-y'(q) dq$ by the nonnegative measure $d(-y(q))$; a downward jump contributes a point mass of its size. The same proof applies, with equality when y is constant on each open ironed interval. Jumps at its endpoints are allowed because the curves agree there. $\square$

Modifying this statement for value space, with $\bar{\varphi}(v) = \bar{\Phi}'(1 - F(v))$:

Theorem 2. *For any monotone allocation rule $x(\cdot)$ and the virtual value function $\varphi(\cdot)$ of F , the expected virtual welfare of an agent is upper-bounded by their expected ironed virtual welfare, i.e.,*

$$\mathbb{E}[\varphi(v)x(v)] \leq \mathbb{E}[\bar{\varphi}(v)x(v)].$$

Furthermore, this inequality holds with equality if the allocation rule x is constant on each value interval corresponding to $\bar{\Phi}(q) > \Phi(q)$, where $q = 1 - F(v)$.

Claim 4. The expected revenue on the ironed revenue curve at any sale probability q is attainable with a DSIC mechanism with that sale probability.

Example: How would you obtain the ironed revenue at quantile q instead of just $P(q)$?

For $q \in [a, b]$ where $\bar{P}(q) > P(q)$ and a, b are the contact endpoints, if $q = \alpha a + (1 - \alpha)b$, we achieve $\bar{P}(q)$ by randomizing the prices $v(a)$ and $v(b)$ with probabilities α and $1 - \alpha$ accordingly to yield $\alpha P(a) + (1 - \alpha)P(b)$ on the concave hull. Equivalently, we mix their sale probabilities a and b with exactly the same weights.

Note: Recall that the expected revenue of *any normalized DSIC mechanism*, not just a posted price, can be expressed by its virtual welfare. (We have now shown that you could decompose a bidder's allocation, with other bids fixed, into a distribution of posted prices and thus express the revenue that way, too, actually.)

What's the final mechanism? Now that $\bar{\varphi}_i(\cdot)$ is monotone (for every i), we choose the $x(\cdot)$ that maximizes $\mathbb{E}_v[\sum_i \bar{\varphi}_i(v_i)x_i(\vec{v})]$, breaking ties by a fixed priority order, which will thus be monotone. Apply the normalized payment identity to obtain a DSIC mechanism. By Theorem 2, this is an *upper bound* on the optimal revenue.

For any ironed interval $[a, b]$ in quantile space, examine $\bar{\varphi}(q)$ for $q \in (a, b)$. $\bar{P}(q)$ is a straight line (linear) there, so $\bar{\varphi}(q)$ will be constant.

What does this imply for ironed-virtual-welfare-maximizing allocation in $[a, b]$? Holding the other bids fixed, it will be constant on (a, b) , and thus its derivative will be zero.

Hence, for this allocation, ironed virtual welfare is equal to virtual welfare by Theorem 2. It attains the upper bound and therefore maximizes expected revenue.

Acknowledgements

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References

- [1] Jason D. Hartline. Mechanism design and approximation. *Book draft. October*, 122, 2013.
- [2] Tim Roughgarden. *Twenty Lectures on Algorithmic Game Theory*. Cambridge University Press, 2016.